

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

In **2010**, a swim club had a total of **35** swimmers, each classified as either advanced or intermediate. From **2010** to **2020**, the number of advanced swimmers in the club increased by approximately **53%**, and the number of intermediate swimmers in the club increased by approximately **44%**. The total number of swimmers in the club increased by approximately **49%**. Which equation best represents this situation, where ***a*** represents the number of advanced swimmers in the club in **2010** and ***b*** represents the number of intermediate swimmers in the club in **2010**?

Answer

- A. $1.53a + 1.49b = 35(1.44)$
- B. $1.49a + 0.53b = 35(1.44)$
- C. $1.53a + 1.44b = 35(1.49)$
- D. $1.44a + 1.53b = 35(1.49)$

Correct Answer: C

Rationale

Choice C is correct. It's given that in **2010**, a swim club had a total of **35** swimmers, each classified as either advanced or intermediate, and that ***a*** represents the number of advanced swimmers in **2010** and ***b*** represents the number of intermediate swimmers in **2010**. It's also given that from **2010** to **2020**, the number of advanced swimmers in the club increased by approximately **53%** and the number of intermediate swimmers in the club increased by approximately **44%**. Thus, in **2020**, the approximate number of advanced swimmers in the club can be represented as **$1.53a$** and the approximate number of intermediate swimmers in the club can be represented as **$1.44b$** . It's given that the total number of swimmers in the club increased by approximately **49%** from **2010** to **2020**. Since the club had **35** swimmers in **2010**, it follows that the total number of swimmers in **2020** can be represented as **$35(1.49)$** . Since the sum of the number of advanced swimmers in **2020** and the number of intermediate swimmers in **2020** equals the total number of swimmers in **2020**, the equation **$1.53a + 1.44b = 35(1.49)$** best represents this situation.

Choice A is incorrect. This equation represents a situation where the number of intermediate swimmers in the club in **2020** increased by approximately **49%**, not **44%**, and the total number of swimmers in the club in **2020** increased by approximately **44%**, not **49%**.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect. This equation represents a situation where the number of advanced swimmers in the club in **2020** increased by approximately **44%**, not **53%**, and the number of intermediate swimmers in the club in **2020** increased by approximately **53%**, not **44%**.